

PSYCHOLOGY

Year 11

MARKING GUIDE 2023

Section One: Short Answer**70% (102 marks)**

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Question 1**Total 25 marks**

- a) Explain whether the study should have a directional or a non-directional hypothesis.

(2 marks)

Description	Marks
Non-directional, the study does not allow the researcher to predict a result.	0-2
Total	2

- b) Explain why using stratified sampling could improve the above study. (2 marks)

Description	Marks
Participants are chosen by certain characteristics and then a sample is taken from this (1) this allows a researcher to represent a population not just year 12s(1)	0-2
Total	2

- c) Calculate the mean values for amount of sleep, and scores on memory test.

(2 marks)

Description	Marks
Amount of sleep: 6.9	0-1
Memory test score: 14.4	0-1
Total	2

- d) Calculate the median values for amount of sleep, and for scores on the memory test.

(2 marks)

Description	Marks
Sleep: 7	0-1
Memory test: 15.5	0-1
Total	2

- e) Identify **one (1)** extraneous variable that the researcher should have controlled. How could this variable affect the study. (3 marks)

Description	Marks
Any one reasonable answer e.g Learning difficulties, sleeping issues 2 marks for how it could affect the study: e.g if a person is unable to remember facts they could have reduced scores	0-3
Total	3

- f) Explain the concept of informed consent and how this would be used in the above experiment. (3 marks)

Description	Marks
- Informed consent is when a participant is given all the information about the experiment and agrees to participate (2) - The students are under the age of 18 so both student and parents need to consent to the experiment (1)	0-3
Total	3

- g) Identify the type of data collection used and state **one** advantage and **one** disadvantage of this type. (3 marks)

Description	Marks
- Subjective Quantitative (1) - Advantage: statistically analyse/numerical data (1) - Disadvantage: open to bias, as participants may not be honest (1)	0-3
Total	3

- h) Using the grid provided, draw an appropriate graph for the data in the table on the previous page.

(6 marks)

Description	Marks
• Correct graphing from table • Title • X axis labelled correctly with units • Vertical axis labelled correctly with units • Y axis appropriate scale • X axis appropriate scale	0-6
Total	6

- i) Describe the relationship of the correlation from the graph.

(2 marks)

Description	Marks
• Strong positive (1) • As number of hours of sleep increases, the score on the memory test also increases (1)	0-2
Total	2

Question 2**Total 12 marks**

Milly was running through the bush when she felt a snake brush past her leg. She immediately stopped and stood still.

- a) Explain the role of sensory neurons, interneurons and motor neurons in Milly's scenario. (6 mark)

Description	Marks
Sensory receptors would detect the snake and sensory neuron will send this information to the brain (2) Interneurons would receive the information in the CNS and transfer it to the motor neuron (2) the motor neuron would send this information to the muscles which would freeze Milly still (2)	0-6
Total	6

- b) Explain what part of the autonomic nervous system would have kicked in once Milly was safe at home, and list **four** psychological responses Milly would then experience. (6 marks)

Description	Marks
Parasympathetic Nervous system her rest and digest response (2) Psychological response: • Decreased heart rate • Decreased breathing rate • Stimulation of digestive system • Constrict pupils	0-6
Total	6

Question 3**Total 11 marks**

a) Fill in the functions of the brain structures in the table below:

(5 marks)

Brain structure	Function
Medulla	Involved in regulating Heart rate, breathing rate and blood pressure
Hypothalamus	Releases hormones and controls homeostasis
Broca's area	Controls speech production
Primary motor cortex	Involved in voluntary movement
Primary Sensory cortex	Involved in sensory activities

Roger Sperry famously researched the brain by using the split-brain experiment.

b) Explain the function of the corpus callosum.

(2 marks)

Description	Marks
Corpus callosum is the band of fibers that connect the two hemispheres together, this allows the hemispheres to communicate between each other. (2)	0-2
Total	2

- c) Identify and explain **two** contemporary methods to improve our knowledge of the structure and function of the brain. (4 marks)

Description	Marks
Any 2 of the following: 1 mark for identify and 1 explanation EEG- measures brain activity by using electrodes to show brain waves. These can detect brain abnormalities such as epilepsy and brain tumors, MRI- using magnets and radio waves to produce a 2D image that can be formed into a 3D picture that can be rotated to look for abnormalities of the brain tissue, fMRI-detects brain function by measuring the amount of oxygen in the neurons of the brain, more oxygen shows more activity of the brain, CT-using x-ray imaging it takes cross-sectional sliced pictures of the brain that can help detect for abnormalities, strokes, and other injuries that may have caused changes in behaviour	0-4
Total	4

Question 4

Total 16 marks

- a) Identify the roles of the following types of brain plasticity in infancy. (2 marks)

Description	Marks
Adaptive Plasticity: the synapses change as we learn	0-1
Developmental Plasticity: synapses change during the foetal and adolescent ages	0-1
Total	2

- b) Identify and explain the stages of plasticity. (10 marks)

Description	Marks
Proliferation: foetal cells that become neurons (2) Migration: neurons move to CNS to determine their functions (2) Circuit formation: axons move outwards and form synapses (2) Synaptic pruning: neurons start to be removed as there are more than needed (2) Myelination: myelin sheath grows around the axon to protect it (2)	0-10
Total	10

Question 5**Total 12 marks**

- a) Bowlby developed a theory on attachment in 1969. Explain the following findings from his research: (2 marks)

- i) Monotropy

Description	Marks
One primary attachment	0-1
Total	1

- ii) Maternal Deprivation

Description	Marks
loss of attachment with mother can lead to delinquency, aggression	0-1
Total	1

- b) State the aim of Harlow's study using rhesus monkeys. (2 marks)

Description	Marks
To determine whether feeding or comfort lead to attachment formation	0-2
Total	2

- c) Explain the findings of Harlow's study using rhesus monkeys. (4 marks)

Description	Marks
Monkeys spent more time with their terrycloth surrogate (that provided comfort) (1) Only went to wire surrogate for feeding (1) Attachment is based not only on physical needs (1) but also having emotional needs met /comfort contact (1)	0-4
Total	4

- d) Identify **two** strengths and **two** limitations of his theory. (4 marks)

Description	Marks
Strengths: helped social workers understand neglect and abuse in children/use of animals to understand attachment in children and recognize those at risk Limitations: ethical criticisms of cruelty to animals/limited value in understanding deprivation in humans	0-4
Total	4

- (a) Explain this attribution according to Heider's theory. (2 marks)

Description	Marks
Dispositional/internal (1) - his behaviour is attributed to his internal characteristics such as his risk-taking attitudes /enjoyment of dangerous sports (1) or Fundamental attribution error/bias (1) - his behaviour has been attributed by others to internal characteristics such as risk-taking/enjoyment of dangerous sports (1)	1–2
Total	2

Name and describe the key concept of the social psychology theory proposed by Leon Festinger. (2 marks)

Description	Marks
Cognitive dissonance theory	1
Tension occurs within an individual when holding conflicting beliefs, attitudes or behaviours	1
Total	2

Apply Festinger's theory to explain why Jos's friends struggle to accept the findings of the police report. (2 marks)

Description	Marks
The friends attribute/believe the accident is caused by Jos's behaviours	1
The police report contradicts their attribution/belief	1
Total	2

Refer to Festinger's theory to explain **two** ways in which Jos's friends could accept the report's findings. (4 marks)

Description	Marks
Two marks for each way	
Answers could include:	
• Jos's friends could change one thought (1) e.g. Jos is careful when driving with other people. (1)	1–2
• Jos's friends could add a new thought to rationalise (1): E.g. Jos was influenced by his friends/the police are the experts (1)	1–2
• Jos's friends could trivialise the inconsistency (1): e.g. It doesn't matter as no-one was hurt (1).	1–2
Total	4
Note: Reference to change of behaviour is not relevant to the scenario. Accept responses that apply to the scenario.	

End of Section Two

Section Three: Extended Answer**30% (48 Marks)****Question 8****21 marks**

Annabelle and Archie are siblings. Annabelle is 8 years old and Archie is 1 years old. When dropping Archie at day care, Annabelle noticed that he was playing with his toys when they left and didn't even say goodbye. Annabelle's mum told her that she was never like that, when Annabelle was picked up she would cling to her mum kicking and screaming.

- Identify and explain the stages of cognitive development the siblings would most likely be in and the corresponding abilities they would have according to Piaget. (6 marks)
- Identify and explain **one** Piagetian task that could be used to place the siblings in the correct stage. (2 marks)
- Identify the attachment types the siblings would have according to Ainsworth. (6 marks)
- Explain, using empirical evidence if patterns of attachment would be similar in different cultures. (4 marks)
- Quality of response. (3 marks)

Question 8 Guide to Marking Extended Responses		Marks
Piaget		6
Annabelle would be in stage Concrete operational stage (1), she would be able to conserve mass and volume (1) and think logically (1) Archie would be in sensorimotor stage (1) he would only view the world through his senses and motor abilities (1) and would not have object permanence yet as he would believe objects disappear when they are out of sight (1)		0-6
Piagetian Task		2
Conservation task: Two glasses task, ask children if the amount is the same after pouring one glass of water into a different shaped container (or plasticine) (2) OR Invisible Displacement: ability to track an object when it has been hidden out of sight.		0-2
Attachment Type		6
Archie would be type A: insecure avoidant: isn't bothered when mother leaves or returns/ ignores mother, Archie doesn't say goodbye Annabelle would be type C: anxious resistant: resists the mother but clings to her and hits her away at the same time. She used to kick and scream and cling to her mum when she came to pick her up		0-6
Attachment Cultures		4
• Van Ijzendoorn and Kroonenberg • Type B was the most common amongst cultures. • Type C most common in Japan and Israel • Type A most common in Western Europe		
Structure		3
A well-constructed answer, use of appropriate psychological language		3
Answers with well-developed sentences and paragraphs		2
Lack of paragraph structure and incoherent responses		1
Answer is too brief or irrelevant		0
TOTAL		/21

Question 9**27 marks**

Isla and Grace were 2 and 4 year-old girls, who's mothers were best friends. When Isla's mum caught her chewing her squeaky giraffe toy, she said "All she does is chew her toys and put them in her mouth like she is eating them."

Grace's mum replied with "I have been through that too. Grace used to chew her blanket all the time and then she realised that it was for comfort not for eating. The phase will stop."

- Explain the process of schema formation in the two girls. (14 marks)
- Identify and explain the stages of physical development that the girls are experiencing. (4 marks)
- Identify and explain the changes in language development that the girls are experiencing. (6 marks)
- Quality of response (3 marks)

Question 9 Guide to Marking Extended Responses		Marks
Schema		14
• Schemas are knowledge learnt through experience and as a child grows these experiences change. (1) • Isla's schema is that everything she holds is edible and goes into the mouth. (1) • The assimilation is new information fitting into existing information (2) is that food goes in her mouth and then her new toy also goes in her mouth because she is holding it (2) • Isla would have reached disequilibrium where she is in conflict about the two objects this stage helps form accommodation/new schemas (2) • Grace's mum was explaining that she went through the same assimilation, but has reached an equilibration where she has reached new cognitive growth in her schemas and has developed an accommodation (2) • Accommodation is when new experiences are different to our existing experiences and we need to change our existing schemas and make new ones (2) • Isla now knows not all things in your hand are to be chewed her blanket is for cuddling and comfort (2)		0-14
Physical Development		0-4
Isla's physical development is from 0-2 she would be finalising her motor skills of sitting, crawling and walking (2) Grace will be moving on to motor skills developing into running, skipping and throwing (2)		0-4
Language Development		0-6
Isla would be developing simple concepts/understanding phrases with support from pictures/ uses between 200-300 words/can hold conversation but will not follow topics (3) Grace can understand questions/aware of time/can hold conversations and turn-taking/uses long sentences (3)		
Structure		3
A well-constructed answer, use of appropriate psychological language		3
Answers with well-developed sentences and paragraphs		2
Lack of paragraph structure and incoherent responses		1
Answer is too brief or irrelevant		0
TOTAL		/27

Question 10**27 marks**

Trevor is a huge adrenaline seeker. He has abseiled and sky dived, and his new favourite thing is to zip line across the canyon. Although he loves the adrenaline rush, he also feels pleasure in doing the things he loves. He finds that his dopamine levels increased dramatically when he is zip lining.

- Explain the parts of the nervous system that would be triggered during Trevor's ziplining experience, and the effects on his body during this process. (18 marks)
- Explain how the dopamine would be transmitted around his body. (5 marks)
- Quality of Response (3 marks)

Question 10 Guide to Marking Extended Responses		Marks
Nervous system		18
• Nervous system composed of Central nervous system the brain and spinal cord (2) • Peripheral NS consists of neurons (2) • Peripheral NS splits into the efferent division which would carry impulses to Trevor's muscle through motor neurons (2) and the afferent division which would carry impulses to the CNS from Trevor's receptors, skin on the ropes (2) • Efferent division is split into somatic NS which is voluntary movements, Trevors muscles moving along the zipline (2) and the autonomic NS which is the involuntary movements would be his HR etc during the zipline (2) • Trevor would then be using the sympathetic NS (fight or flight response) preparing his body for strenuous activity (2) • His heart rate would increase (1) his bronchi would dilate (1) his pupils would dilate (1) liver production would increase glucose (1)		0-18
Neurotransmission		6
• Action potential reaches axon terminals • Neurotransmitters/dopamine (chemical messengers) released from vesicles in axon into synapse • synapse is the gap between neurons • Neurotransmitters are collected from synapse into receptors in receiving dendrite • Leftover neurotransmitters are destroyed in the synapse by enzymes • Leftover neurotransmitters Reuptake into the axon		0-6
Structure		3
A well-constructed answer, use of appropriate psychological language		3
Answers with well-developed sentences and paragraphs		2
Lack of paragraph structure and incoherent responses		1
Answer is too brief or irrelevant		0
TOTAL		/27

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Acknowledgements

**Information from: Fletcher, J., & Garton, A. (2017). *Psychology – Self and Others*.
Sydney: Pearson education Australia**

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